THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL UNIT TEST SET 3

Chapters 5–8: **Describing Motion Around Us, How Forces Affect Motion, Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A 100 W bulb and a 2 kW heater are used for 3 hours and 1 hour respectively. The ratio of energy consumed is: 1

(a) 1 : 20 **(b)** 3 : 2 **(c)** 3 : 20 **(d)** 20 : 3

2. A pulse takes 0.1 s to travel along a 34 m long steel pipe. The speed in steel is: 1

(a) 3400 m/s **(b)** 340 m/s **(c)** 0.0029 m/s **(d)** 34 m/s

3. A body covers the first half of its journey at 20 m/s and the second half at 30 m/s. The average speed is: 1

(a) 24 m/s **(b)** 26 m/s **(c)** 50 m/s **(d)** 25 m/s

4. Assertion (A): A heavier object is harder to set in motion than a lighter one. Reason (R): A heavier object has more inertia. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. Assertion (A): A body thrown up has maximum potential energy at the highest point. Reason (R): The kinetic energy is maximum there. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. Assertion (A): A louder sound has a greater frequency. Reason (R): Loudness depends on the amplitude only. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. Assertion (A): A ball thrown vertically up has zero acceleration at the highest point. Reason (R): Its velocity is zero at the highest point. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. A balloon moving upward at constant velocity releases a stone. Just after release, the stone's acceleration is: 1

(a) 10 m/s squared downward **(b)** 10 m/s squared upward
(c) zero **(d)** 20 m/s squared

9. The ratio of KE of two bodies of masses 1 kg and 4 kg with equal momentum is: 1

(a) 2 : 1 **(b)** 4 : 1 **(c)** 1 : 4 **(d)** 1 : 1

10. A person hears the echo of a gun shot 1 s after the shot and a second echo 3 s after. The ratio of the distances of the two reflectors is: 1

(a) 3 : 1 **(b)** 1 : 1 **(c)** 1 : 3 **(d)** 1 : 2

11. The ratio of distances covered in the 1st, 2nd and 3rd seconds from rest with uniform acceleration is: 1

(a) 1 : 3 : 5 **(b)** 1 : 2 : 3 **(c)** 1 : 4 : 9 **(d)** 1 : 1 : 1

12. A car and a truck, both with the same momentum, are brought to rest in the same time. The retarding force is: 1

(a) the same for both **(b)** zero
(c) greater for the truck **(d)** greater for the car

13. **Assertion (A):** The joule is the SI unit of energy. 1
Reason (R): Power is the rate of doing work.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** Sound waves in air are longitudinal waves. 1
Reason (R): The speed of sound is greater in solids than in gases.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** Acceleration is the rate of change of velocity with time. 1
Reason (R): The acceleration of a body is always in the direction of its motion.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why does the PE of a body not depend on the path taken to reach a height? 2

17. State why the sound of an approaching train is first heard through the rails. 2

18. A ball is thrown vertically upward and returns to the thrower in 4 s. Find the speed of throw and the maximum height (g = 10). 2

19. Two forces of 9 N and 5 N act on a 2 kg body in the same direction and then in opposite directions. Find the accelerations. 2

20. A 1 kg ball is thrown vertically up with a KE of 100 J and rises to only 8 m. Find the energy lost to air resistance (g = 10). 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. A 60 kg man climbs a 5 m high ladder in 10 s carrying a 20 kg load. Find the work done and the average power (g = 10). 3

22. A man fires a gun in front of a cliff and hears an echo after 2 s and a second after 3 s from another cliff. Find the distance between the cliffs if he stands between them (v = 340 m/s). 3

23. A man jumps from a height of 5 m. Find the time of fall and the speed on landing, and the retardation if he stops in 0.1 s (g = 10). 3

24. A girl pushes a box with 40 N and the friction is 40 N. Describe the motion and explain what happens when she pushes with 60 N. 3

OR

A motor lifts a 200 kg elevator through 30 m at a constant speed in 20 s with 60 per cent efficiency. Find the input power (g = 10). 3

25. A man claps near a big cliff and hears the echo after 3 s. He then walks 85 m toward the cliff and claps again. Find the new echo time (v = 340 m/s). 3

OR

A car travelling at 25 m/s is overtaking another at 15 m/s on a road. A truck coming from the opposite direction at 20 m/s is 400 m ahead. Will the overtaking car pass safely if it needs 6 s? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** Ravi, a student of mass 50 kg, climbs the stairs of his school building through a vertical height of 10 m in 20 s. Take $g = 10$ m/s^2. 5

(i) Find the gain in his potential energy.
(ii) How much work does he do against gravity?
(iii) Find his power.
(iv) If he climbs the same stairs in 40 s, what is his power, and does the work done change?

27. A car of mass 1200 kg accelerates from rest at 1.5 m/s squared for 10 s against a constant resistive force of 600 N. Find 5

(a) the final speed
(b) the KE
(c) the work done by the engine
(d) the average power
(e) the final power.

OR

Describe an experiment to measure the speed of sound in air using two observers and a gun or a clapper at a large distance. State the observations, the calculation for 1020 m with a time of 3 s, and the precautions. 5

END OF QUESTION PAPER — **Total: 50 Marks**